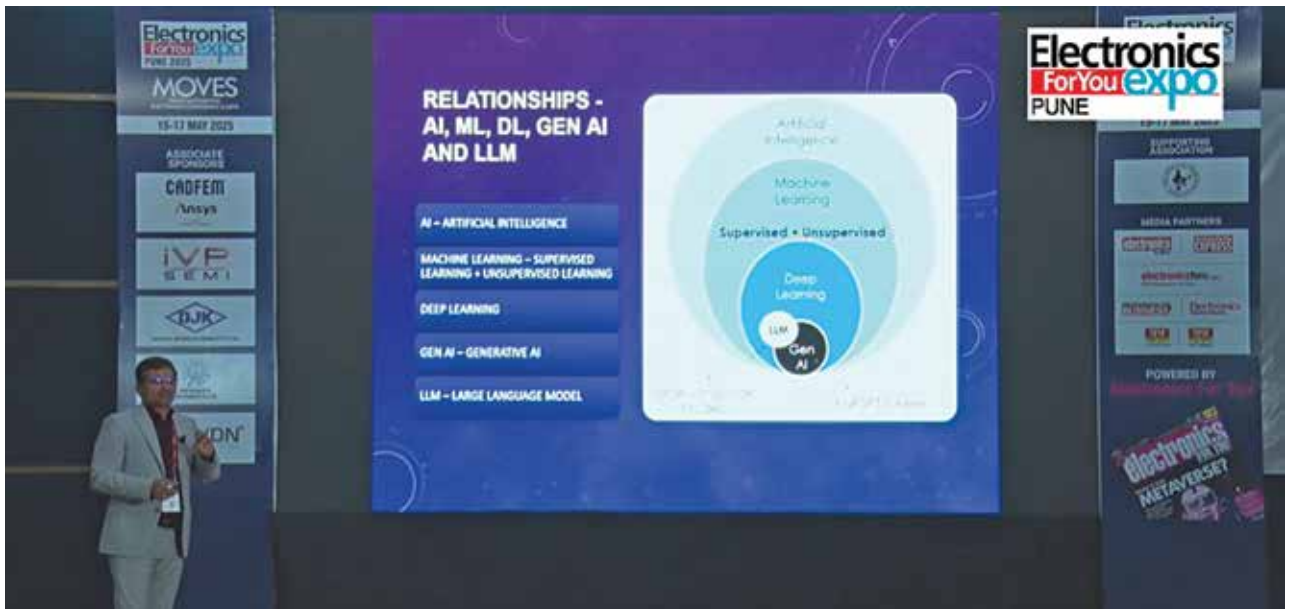


“The AI revolution is here —and you are not too late to join”

From robot-run hospitals to AI-written code, AI is here, and it is learning faster than most of us can imagine. How can a beginner in this field catch up with this? **Abhijeet Desai** from **HCLTech America** drew a roadmap in his engaging session at the EFY Expo in Pune.



Artificial intelligence (AI) has evolved significantly from early concepts to today’s generative tools. By 2035, over 75 billion devices are projected to be connected globally, underscoring the explosive growth of the Internet of Things (IoT). In parallel, the AI market is expected to reach \$2 trillion by 2030.

These estimates are based on insights from respected firms like McKinsey, which derive their findings from extensive interviews with global business leaders—CTOs, CIOs, CFOs, and CEOs—across various industries.

To grasp AI’s trajectory, it helps to consider its impact across three timeframes: short, medium, and long term. In the short term, AI will drive productivity—much like automation, computing, and software did in previous

decades. The medium term will focus on complete automation and sharper efficiency. In the long term, AI will introduce greater complexity and fuel demand for increasingly intelligent systems. Tools like ChatGPT may seem complex today, but they will likely continue to grow more advanced.

But, despite concerns, AI is not here to replace all jobs. Instead, it is prompting companies like Microsoft, IBM, and Google to restructure and invest in AI infrastructure and talent. This is part of the natural business cycle.

Smart devices are becoming ubiquitous, ranging from intelligent shoes to showers. Hardware intelligence and embedded systems are evolving rapidly, supported by technologies such as augmented reality (AR), chip design, and advanced manufacturing. In China,

there is a hospital that operates entirely with robots, eliminating the need for human intervention. In the US, banks like Bank of America are reducing the number of physical branches due to AI-driven innovations.

Real-time systems and edge computing are also transforming development. Instead of sending all data to the cloud, processing at the edge saves time and resources. Business logic and data remain local, with only critical, aggregated insights transmitted to the cloud.

The demand for quick, cost-effective innovation has made edge computing and smart IoT essential in Industry 4.0. As roles evolve, continuous learning remains key. It is essential to remain curious and adapt to the evolving technological landscape.